the amount of money spent on it in the United States, where the rough structure of a giant cyclotron was completed in 1940; its size makes it look more like a battleship than a scientific instrument. Its pole pieces are 4·7 metres in diameter, and its magnet is 17·8 metres in length (Figure 39). Its foundation contains 1,200 tons of concrete; the magnet contains 3,700 tons of iron and 300 tons of copper, wound in a strip of 10·2 cm. in width and 6 mm. in thickness. The frame of the magnet is made of 36 steel plates, each 5·5 mm. in thickness. The magnetic field intensity is 10,000 oersteds, and the frequency of the alternating electric field corresponds to a wavelength of 39 metres. Lawrence completed this cyclotron after the war, and it has enabled him to accelerate deuterons up to 100 Mev. and alpha particles up to 200 Mev.

In other words, the cyclotron is an extremely costly and also extremely complex apparatus. However, it is still by far the most useful nuclear-physical research instrument designed for the same purpose. In the United States it has made it possible to accomplish many nuclear reactions which would not have been feasible by any other means.

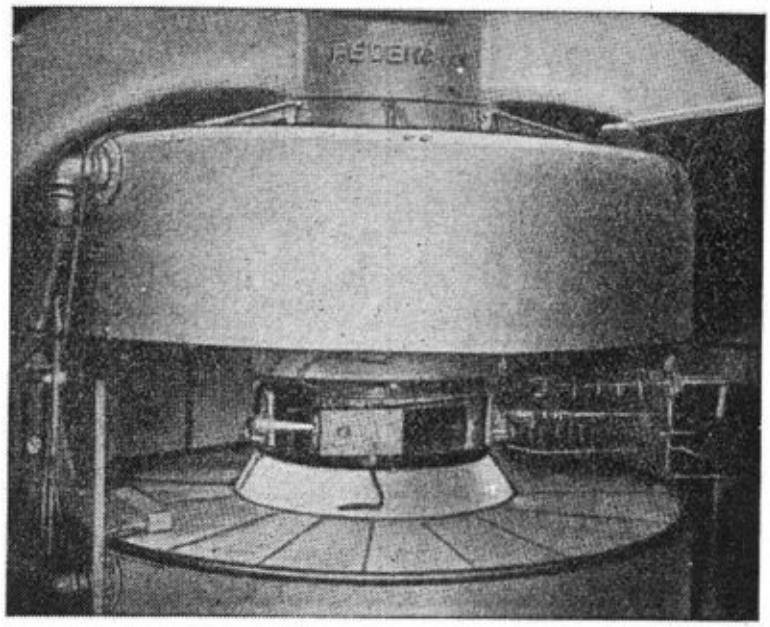


Figure 38.—Cyclotron.